Bismuth: A Peer-to-Peer Settlement Layer for Programmable Value

The Bismuth Foundation

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Abstract. A purely peer-to-peer settlement layer would let value move, and programs execute, directly between parties without any intermediary taking custody. Proof of work answers who may extend the ledger; it does not answer what a transaction is allowed to express. Bismuth keeps one account-based ledger and one memory-hard proof of work, and makes the transaction an abstract, extensible object: a payment carries only the economic fields it needs, while an optional operation and an optional data field let the same record also express a token, a name, a contract call, or a private transfer — every class ordered, validated, and committed through one consensus path, with no special-purpose block types. A transaction is identified by the hash of its own content; signatures are verified through a family-agnostic framework spanning classical and post-quantum schemes; a deterministic RISC-V virtual machine custodies and moves real value under a state root the network commits to; a consensus-validated shielded layer carries private transfers; and state is one canonical chain plus rebuildable key-value projections. This protocol is brought into force by a single miner-activated upgrade, hf2, whose activation status on the public network is stated in Section 16; as of this edition hf2 has not activated on the public network, so the capabilities above are not yet live there. This document describes that system and presents the disclosures for the BIS crypto-asset on the template of Regulation (EU) 2023/1114 (Markets in Crypto-Assets, "MiCA"), Article 6 and Annex I.

STATEMENT UNDER ARTICLE 6(3) OF REGULATION (EU) 2023/1114

This crypto-asset white paper has not been approved by any competent authority in any Member State of the European Union. The Bismuth Foundation, as the person that drew up this crypto-asset white paper, is solely responsible for the content of this crypto-asset white paper.

STATEMENT OF THE MANAGEMENT BODY (ARTICLE 6(6))

This crypto-asset white paper complies with Title II of Regulation (EU) 2023/1114 and, to the best of the knowledge of the management body of the Bismuth Foundation, the information presented in this crypto-asset white paper is fair, clear and not misleading and this crypto-asset white paper makes no omission likely to affect its import.

Summary

> This summary is provided under Article 6(7) of Regulation (EU) 2023/1114. It should be read as an introduction to this crypto-asset white paper. Any decision to purchase BIS should be based on the content of the white paper as a whole and not on this summary alone. The offer of a crypto-asset, where one is made, does not constitute an offer or solicitation to purchase financial instruments, and any such offer or solicitation can be made only by means of a prospectus or other offer documents pursuant to applicable national law. This crypto-asset white paper does not constitute a prospectus as referred to in Regulation (EU) 2017/1129 or any other offer document pursuant to Union or national law.

Bismuth is a public, permissionless, proof-of-work blockchain first mined on 1 May 2017. Its native crypto-asset is BIS. There was no pre-mine, no founder allocation, and no initial coin offering; all BIS is created by the protocol as a block reward. Most of each block's reward accrues to the miner that finds the block; in addition, on every tenth block until block 4,380,000, the protocol paid a development reward equal to that block's mining reward, and a separate, declining hypernode reward, to fixed protocol-designated addresses; the chain has since passed that height and those periodic payments have ended (see Section 17). The network is run by independent nodes and miners; no person controls settlement or the order book.

The ledger is account-based. A transaction is an eight-field record — timestamp, sender, recipient, amount, signature, public key, an *operation* string, and an *openfield* data field. The last two fields are a deliberate extension channel: tokens, names, contract calls, and private transfers are expressed through them and are ordered, validated, and committed through the same consensus path as an ordinary payment, rather than fragmenting the chain into many special-purpose transaction types. Proof of work is Heavy3, a memory-hard function that forces each hash attempt to read from a one-gibibyte table, blunting the advantage of specialized hardware over commodity machines.

Bismuth verifies signatures through an abstraction that already supports several cryptographic families — RSA (the scheme in dominant use on the public network today), secp256k1, Ed25519, secp256r1, lattice-based ML-DSA, and native multisignature (the last three activate as spendable sender families only at hf2) — selected by the sender's address rather than hard-coded into consensus, which positions the protocol to adopt post-quantum signatures without redefining how it thinks about identity. A deterministic RISC-V virtual machine that lets contracts custody and move real BIS, and a consensus-validated shielded layer that carries private transfers inside ordinary transactions, take effect at the *hf2* upgrade. As of the date of this document the public network has not activated hf2, so a holder should not assume those capabilities, or the other hf2 features, are live on the public network yet; their status is set out in Section 16.

BIS confers no claim against any person and grants no right to income, dividend, redemption, or governance. It is a volatile crypto-asset whose supply has no fixed maximum (Part G); its environmental footprint as a proof-of-work asset is disclosed in Section 18. The disclosures required for a crypto-asset other than an asset-referenced token or e-money token follow in the parts mapped to Annex I of MiCA, and the risk factors are set out in Part I.

CLEAR AND UNAMBIGUOUS STATEMENT (ARTICLE 6(5))

Prospective holders are warned that:

a. the crypto-asset may lose its value in part or in full;
b. the crypto-asset may not always be transferable;
c. the crypto-asset may not be liquid;
d. BIS is not a utility token and is not offered against the promise of any good or service; it confers no claim on the Bismuth Foundation or on any other person;
e. the crypto-asset is not covered by the investor compensation schemes under Directive 97/9/EC;
f. the crypto-asset is not covered by the deposit guarantee schemes under Directive 2014/49/EU.

This crypto-asset white paper contains no assertions as to the future value of the crypto-asset, other than the statement above (Article 6(4)).

> ***Regulatory positioning.*** Bismuth is a fully decentralized proof-of-work crypto-asset with no issuer and no offeror. BIS is created only as a reward for validating transactions and maintaining the distributed ledger; it is not sold by, and confers no claim on, any person. In the Bismuth Foundation's good-faith assessment, under Recital 22 and Article 4 of MiCA the public distribution of a crypto-asset that is created as a mining reward and offered for free does not require a crypto-asset white paper; a competent authority could nonetheless assess the regulatory characterisation of BIS, or the standing of this document, differently. The white-paper obligation under Article 5 is engaged only on admission to trading, and it then falls on the person seeking admission to trading or on the operator of the trading platform (Article 5(3)), not on a development foundation.
>
> The Bismuth Foundation is the core development steward of the protocol. It is not an issuer, has not conducted an offer to the public, and makes no MiCA filing by publishing this document. This paper is published voluntarily, drawn up on the template of Article 6 and Annex I, so that holders, prospective holders, and any person who may seek admission of BIS to trading have the comparable, structured, risk-aware information that MiCA is designed to provide. Where Annex I assigns a disclosure to an "issuer" or "offeror," that role does not exist for Bismuth and the item is completed accordingly. Section 22 records the document's standing in full.

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1. Introduction

A public blockchain has to answer two questions that are easy to confuse. The first is *who* may extend the ledger, and in what order; the second is *what* a transaction is allowed to express. Most designs answer the first with a consensus rule and then bolt the second onto it, growing a thicket of special-purpose transaction types, side systems, and execution environments, each with its own validity rules, its own failure modes, and its own attack surface. The base layer becomes large, and large base layers are hard to reason about and to maintain.

Bismuth takes the opposite discipline. It answers the first question with proof of work and a single account-based ledger, and it answers the second by making the transaction an abstract object with two optional extension fields — a free-form *operation* string and *openfield*. Tokens, names, contract calls, and private transfers are not new kinds of block: they are ordinary transactions whose operation and openfield carry the payload, ordered, validated, propagated, and committed through one consensus path. The base node need not understand any of them; the consensus engine stays small and explicit, while the meaning that can ride on top of it stays open-ended.

This document describes the protocol in the form defined by its coordinated upgrade, *hf2*, which folds the modernization of serialization, identity, proof of work, difficulty, fees, and execution into a single activation boundary. That is the protocol Bismuth is built to be, and the description that follows is in the present tense throughout; the activation mechanism and the current status of *hf2* on the public network are set out in Section 16 and in the risk disclosures of Part I. Consensus byte formats are defined in one place and locked by characterization tests, so that a change to storage, transport, or wallet code cannot silently alter what the chain means. The remainder of the paper describes the ledger and its transaction model (Sections 2 and 6–9), the proof of work and consensus (Sections 3–5), the execution and privacy layers (Sections 10–11), the storage, network, and wallet stack (Sections 12–14), the cryptographic roadmap (Section 15), the upgrade boundary (Section 16), and issuance (Section 17), after which the crypto-asset disclosures required by Regulation (EU) 2023/1114 follow in the parts mapped to its Annex I.

2. The account ledger and the abstract transaction

The ledger is account-based: the protocol tracks a balance for each address rather than a set of unspent outputs. A transaction names a sender, a recipient, an amount, a signature, and a public key or key reference, and it carries the two extension fields. On the wire it is the tuple

```
timestamp | sender | recipient | amount | signature | public_key | operation |
openfield
```

and the committed ledger row adds the block height, the block hash, the fee, and the reward. Amounts are denominated in units of 10^{-8} BIS — eight decimal places, the smallest unit being one hundred-millionth of a BIS — and every amount is normalized to that precision before it is hashed or checked. In the canonical bytes an amount is an integer count of those atomic units (Section 6).

The sender's balance is authoritative. When a block is validated, the node does not check each transaction in isolation; it sums every debit and fee that each address incurs across the *whole* candidate block and rejects the block unless each sender can cover its total. This block-cumulative rule is what prevents an address from spending the same balance twice through several transactions in one block, and it is enforced before the block is committed.

The `operation` and `openfield` fields are the heart of the design. `operation` is a free-form namespaced command — for example `token:issue`, `token:transfer`, `alias:register`, `vm:deploy`, `vm:call`, or `shield:spend` — and `openfield` carries the associated data. The base node orders and archives these transactions without needing to understand them; the subsystems that do understand them — the token and alias indexes, the virtual machine, the shielded pool — read the same canonical rows. Consensus validation of `vm:` and `shield:` operations activates at hf2 (Sections 10, 11, and 16). A protocol can therefore be added at the interpretation layer without a new block format and without enlarging the consensus core. Both extension fields are optional: an ordinary payment carries neither, and an empty operation or openfield is dropped from the canonical transaction bytes entirely rather than encoded as an empty field (Section 6). The eight slots in the tuple above are the wire and ledger shape of a transaction; the bytes it actually commits to — the content it signs and is identified by — exclude the signature and public key and include only the extension fields it uses.

3. Proof of work: Heavy3

Bismuth secures block production with Heavy3, a memory-hard proof of work. The cost of a hash attempt is dominated not by arithmetic but by random access into a deterministic one-gibibyte table, `heavy3a.bin`, generated from an HMAC-DRBG seeded with a fixed description of the element bismuth and verified against sentinel values on load. Because the table is large and addressed pseudo-randomly, a miner must hold all of it in fast memory; because it is deterministic, every node can regenerate it. The intent is to keep the marginal advantage of specialized hardware over commodity GPUs small, so that the work stays broadly reproducible.

A mining attempt forms a preimage from the miner's address, a nonce, and the previous block hash, and hashes it to a 224-bit value. That value is then *annealed*: its seven 32-bit words are exclusive-ORed with seven consecutive words read from `heavy3a.bin` beginning at an offset the hash itself selects. Difficulty is measured by an unusual but fixed rule — the length of the longest leading bit-prefix of the target derived from the previous block hash that appears anywhere as a substring of the annealed value's bit-string. The measure does not change the probability of finding a good candidate and is verified by every node identically.

The inner hash is BLAKE2b truncated to 28 bytes. The annealing step, the lookup table, and the substring measure do not depend on that choice, so the digest primitive is the only part the protocol modernizes here. Before the hf2 activation the inner hash is SHA-224: the network mines and validates the SHA-224 form below the activation height and the BLAKE2b form at and above it, with everything else about the work identical (Section 16).

4. Difficulty and block timing

The chain targets a sixty-second block interval. Difficulty is retargeted by an LWMA rule: a linearly weighted moving average of recent solve times whose adjustment is symmetric in the logarithmic work domain, so the chain responds to sustained fast and sustained slow blocks under the same law. It clamps individual solve times to resist timestamp manipulation, bounds the per-step change, and computes entirely in fixed-precision decimal arithmetic so that every platform derives an identical target. The earlier difficulty rule was a feedback controller over a 1,440-block window, asymmetric — quick to raise difficulty and slow to lower it — which punished a small, bursty network when transient hashpower arrived and then left; the LWMA, over a 60-block window, is the symmetric replacement. That earlier controller was the subject of peer-reviewed control-systems analysis

[2, 3], which is cited among the prior-art "Other Publications" of IBM's patent US 10,880,073, *Optimizing Performance of a Blockchain* (granted 2020) [15].

As an operational safeguard — not a consensus rule — a node also runs a background detector that compares its local difficulty against the median of height-matched peers and can schedule a rollback-and-resync if it finds itself trapped on a wrong difficulty track from corrupted local state.

5. Consensus pipeline and chain selection

Every block enters the chain through one admission routine. The node rejects banned or hostile peers, takes the ledger lock, parses and normalizes each transaction into its consensus form, checks signatures, enforces timestamp and replay rules, recomputes the target difficulty, rebuilds the block hash from the canonical representation, validates the Heavy3 proof of work, enforces the block-cumulative balance and fee rules, applies the post-fork feature gates where they are active, commits the block atomically, removes the confirmed transactions from the mempool, and advances the auxiliary indexes. Proof of work is checked only after the transaction list has been normalized, because both the block hash and the mining target are defined over that normalized list.

The network follows the longest valid chain. A node tracks the tip height that its peers report, weighs those opinions by a local reputation score earned from delivering valid data, and synchronizes toward the greatest height that reputable peers agree on; at equal height the competing block hashes decide. Reputation never overrides validation — every candidate block is independently checked for proof of work, difficulty, signatures, replay, and balances before it is accepted — it only hardens the choice of which peer view to trust when recovering from a disputed tip. A cumulative proof-of-work total is well defined over the chain — but it is deliberately not the fork-choice rule, which is height, with the block hash distinguishing competing blocks at equal height. Validation failure at any stage rejects the block, penalizes the peer, and restores the tip from authoritative state; when a rollback occurs the chain rewinds and every auxiliary store rewinds or rebuilds with it.

6. Transaction identity and serialization

A transaction is identified by the hash of its own content. The identifier is a BLAKE2b hash of a canonical preimage that excludes the signature and public key and omits an empty operation or openfield, so a transaction commits to exactly the fields it uses and its identity does not depend on the incidental shape of an encoded signature. The block hash is a BLAKE2b hash over a binary, integer representation of the block's transactions, chained to the parent block hash, in which timestamps are integer centiseconds, amounts are integer atomic units, and every variable-length field is explicitly length-prefixed — a form in which two implementations cannot diverge on quoting or rounding.

Continuity across the boundary is preserved by height: the dispatcher selects the byte form by destination block height, so a historical block remains byte-identical when replayed while a block at or after the activation height hashes under the content-addressed rules. There is no global mode switch that could reinterpret old history (Section 16).

7. Signatures and the signer abstraction

Bismuth does not hard-code one signature family into consensus. Verification is routed through a signer abstraction in which the sender's address selects the signer class, and the validator dispatches accordingly. The abstraction already implements RSA, secp256k1 ECDSA, Ed25519, NIST P-256 (secp256r1), the lattice-based ML-DSA family (FIPS 204, parameter sets 44, 65, and 87), and native multisignature. Any of these families may originate a spend, and the validator dispatches verification according to the sender's address family. The dominant scheme today is RSA: a 4096-bit key with PKCS #1 v1.5 signatures, whose address is the SHA-224 hash of the public-key PEM rendered as 56 hexadecimal characters. The newer ECDSA and Ed25519 addresses use Base58Check encodings with a `Bis1` prefix, and native multisignature addresses use a `Bism` prefix.

For ordinary single-signature secp256k1 senders, the protocol uses a recoverable model: the sender signs the canonical content identifier of the transaction, the signature is stored as a 65-byte recoverable encoding, the public key is omitted, and the validator reconstructs the signing key from the signature and message and derives the address from it. Low-s canonicality is enforced on this path by rejection: a signature whose s value is zero or exceeds half the group order is refused rather than rewritten, so the stored encoding is non-malleable. Other families keep an explicit or by-reference public key where recovery is impossible or undesirable. The unifying property is that the chain always validates against one authoritative content preimage, while the wire representation of key material may differ by family — which is also what positions the protocol to add a post-quantum scheme as a dispatch-and-encoding question rather than a redesign (Section 15).

8. Issuance, the coinbase, and reward maturity

New BIS enters circulation through the coinbase transaction, the block's reward-bearing record, appended as the final transaction in each block. There is no other issuance path: no pre-mine, no founder allocation, and no sale created any BIS. The coinbase is authorized by the proof-of-work and reward rules, not by a sender signature, and the protocol recomputes the reward from a fixed formula rather than trusting the amount written into the transaction, which is designed to prevent minting value by misreporting it.

The coinbase is compact: because it is authorized by proof of work rather than a signed spend, its signature and public-key slots are not needed for a signature and are repurposed as a mining header — the freed signature slot carries the proof-of-work nonce and the freed public-key slot carries the network's commitment to the virtual-machine state root (Section 10), both bound into the block hash so a miner cannot alter them without invalidating the block. Because the mining header now occupies those slots, the coinbase's `operation` and `openfield` carry no mandatory payload: both are free-form, optional, and uncapped, left empty or carrying arbitrary miner data. A freshly mined reward is also subject to maturity. Because an account-based ledger has no isolated output to lock, the protocol deducts immature reward from a miner's spendable balance until the reward is buried by a fixed maturity window — 100 blocks on the public network, 2 on the development network — so that a deep reorganization cannot retroactively invalidate payments funded by a reward it erases.

9. Fees

Transaction fees are deliberately simple and are paid by the sender to the miner of the block. The base fee is responsive to load: the node derives a multiplier from the recent weight of blocks — a measure of structural load that counts payload size, not just transaction count — clamps it within a bounded range, and applies it to a small baseline, with additional surcharges for operation classes such as virtual-machine and shielded transactions whose validation burden is higher and for the size of the `openfield` payload. The result is a deterministic, congestion-sensitive floor rather than an open-ended fee market, and it discourages using the chain as bulk storage while leaving ordinary payments inexpensive.

10. The virtual machine

Bismuth's contract layer is a deterministic RISC-V virtual machine implementing the RV32I integer instruction set, metered by a per-call gas limit and exposing a small system-call interface for storage, the caller and call value, the block number, hashing, value transfer, and calls into other contracts. The design refuses the premise that each richer application needs its own execution model; instead one instruction set and one state-transition engine host higher-order applications that compile or assemble to it. Contracts are introduced by `vm:deploy` transactions and invoked by `vm:call`; a contract's address is derived from the content identifier of its deploy transaction, so contract identity depends on transaction content rather than on a malleable signature.

The defining property is that a contract can custody and move real BIS. Value sent to a contract is transferred into a dedicated, unspendable custody sink on the ledger and mirrored as a contract balance in the virtual-machine state; when a contract pays out, the node debits that balance and writes deterministic settlement rows from the sink to the recipient. Because custody balances live in the contract state and are reconstructed by replaying the calls, a rollback rebuilds them exactly. Under hf2 the virtual-machine state root — a deterministic commitment over contract code, storage, and custody balances — becomes a consensus object: the miner commits the pre-state root into the coinbase header and a node rejects a block whose locally derived state root does not match, so an implementation bug surfaces as a visible block rejection rather than silent divergence. The virtual machine and its committed state root activate at hf2 (Section 16).

11. Shielded value

The shielded layer is a consensus-validated privacy pool carried inside ordinary transactions, not a second chain. Its transfers are Bismuth transactions whose `shield:*` operations carry notes, ring members, key images, and commitments; their transparent envelope gives them a place in the total order while their inner cryptographic content determines private ownership and spend validity. The shielded state — notes, height-indexed rollback entries, key images, and flow accounting — lives in a dedicated key-value sidecar that rewinds or rebuilds to the chain tip.

The design is staged. The first stage constructs stealth-address outputs so that only the intended recipient can recognize and spend a note while observers cannot link it to the recipient's published shielded address. The second stage spends with linkable ring signatures and key images, so the chain prevents double spends — a key image may appear once — without revealing which ring member was spent. The third stage hides amounts with RingCT: Pedersen commitments, logarithmic Bulletproof range proofs, and a multilayer ring signature, built over the secp256k1 curve. The shielded layer replaces an earlier off-chain prototype whose spent-set logic lived outside consensus; here, validity is enforced by every full node as a consensus rule. Like the virtual machine, the shielded layer is gated on hf2 activation (Section 16).

12. Storage architecture

Storage separates canonical chain data from rebuildable projections. The chain itself is the only authoritative history; balances, transaction-identifier indexes, contract state, shielded notes, reward accounting, token balances, and alias ownership are materialized views that must be reconstructible from the chain and removable on rollback. The keystone is an append-only block store that holds block bodies by height in a key-value engine (LMDB, behind an engine-agnostic seam that also supports a portable SQLite-backed key-value fallback with identical ordering), around which sit key-value projections for balances, identifiers, rewards, contract state, shielded notes, and tokens.

These stores are introduced conservatively. Each is written as an additive shadow alongside the older path and becomes authoritative only after replay verification re-derives the canonical block hashes from chain data and parity checks confirm it means exactly the same thing. The predecessor store of record, which the public network still uses, is a set of SQLite databases — an authoritative full ledger database, a hyperblock database that keeps recent blocks in full while compacting deep history into per-address balance snapshots, and an index database for derived data; the key-value stores run alongside it as shadows and become the read path as the migration completes (Section 16).

13. Networking and synchronization

The native peer-to-peer protocol frames each message as a JSON payload behind a fixed-width length header over TCP. The public network listens on port 5658, the test network on 2829, and the development network on 3030. Peers exchange version information, peer lists, synchronization triggers, heights, mempool contents, and freshly mined blocks. Peer management persists peer sets, limits retries, tracks bans and warnings, records peer opinions of the tip height, and maintains the reputation score that hardens tip selection.

Alongside the socket protocol the node exposes a read-mostly HTTP interface (port 5659 where enabled) that serves the same chain through a more parallel transport: status, difficulty, balances, blocks and ranges, headers, peers, fee and contract state, and an optional transaction-submission endpoint. The HTTP interface does not create a second validation path — submitted transactions pass through the same mempool checks as socket transactions — but it does enable headers-first and range-based fetches and capability discovery for faster sync. A node can also serve and fetch a verified ledger snapshot with a height, size, and checksum manifest, so a new node can install a snapshot and then validate forward from its boundary, reducing reliance on a single bootstrap host while preserving full verification from the snapshot onward.

14. Wallets, hierarchical keys, and multisignature

Key management is a client concern: the chain cares only about addresses and signatures, not how a key was derived, so wallet features need no consensus change. Bismuth supports hierarchical-deterministic wallets in the BIP32 style with BIP39 mnemonic seeds, which derive many addresses from one secret along standard derivation paths, and native multisignature, in which an M-of-N address is derived from a threshold and a sorted set of owner keys, the redeem information rides in the transaction, and the node verifies that a sufficient threshold of distinct owners authorized the spend; spending from a native-multisignature address is a post-fork sender path that activates at hf2 (Section 16), while receiving into one is valid before activation. A multisignature vault can equally be expressed as a virtual-machine contract when programmable approval policy is wanted. A keyless browser provider exposes the wallet to web applications over an origin-isolated message channel without surfacing key material to the page.

15. Post-quantum readiness

Because the signer abstraction selects a verification family from the sender's address rather than fixing one in consensus, adopting a new signature scheme is principally a matter of address dispatch and wire encoding rather than a redesign of how the protocol thinks about identity. The repository already carries implementations of the lattice-based ML-DSA family (FIPS 204), with addresses that commit to the large public key by hash. This does not mean the network has migrated to post-quantum signatures; it means the groundwork is present, and a future transition is far lighter than it would be for a protocol welded permanently to one classical scheme. The signature schemes accepted by public-network consensus today — RSA, secp256k1 ECDSA, and Ed25519 — are all classical; ML-DSA is one of the base-layer sender families whose acceptance is governed by the single hf2 activation boundary (Section 16).

16. The hf2 upgrade boundary

The protocol described in this paper is brought into force at one activation boundary, hf2, rather than through a stream of independent toggles. The serialization, signature, proof-of-work, difficulty, fee, and execution changes are coupled and would be unsafe to switch on piecemeal, so a single upgrade activates them together. Activation is miner-driven and computed from the chain itself: upgraded miners place a readiness marker in their coinbase, and when a window of 1,000 consecutive blocks all carry it, the upgrade locks in at the earliest height that closes such a window and activates at the next multiple of 1,000 blocks above a 30-block burial margin. The lock-in height is a deterministic function of confirmed history, identical on every node and independent of the current tip, and it is persisted alongside the ledger so a restart or resync replays the same height. There is no off-chain vote and no second signal: one marker governs the whole bundle.

As of the date of this document, hf2 has not locked in or activated on the public network, and until it does the public network runs the predecessor ruleset: SHA-224 Heavy3, the feedback-controller difficulty, signature-derived transaction identifiers, text-form signing and block hashing, the fixed base fee, and SQLite as the store of record. The hf2 features — BLAKE2b Heavy3, LWMA difficulty, binary and integer serialization, content-addressed identity, recoverable secp256k1 signatures, coinbase compaction and reward maturity, the responsive base fee, the committed virtual-machine state root, the virtual machine, the shielded layer, key-value storage as the read path, and the base-layer sender families a pre-fork node refuses (native-multisignature, NIST P-256, and ML-DSA senders) — are implemented and exercised on the development network and in the test suite, but a holder should not assume any of them is live on the public network before activation. Activation depends on miner adoption; it may be delayed and is not guaranteed (Part I). This is the single most important operational fact for reading the rest of this document.

17. Supply and the reward schedule

BIS was first mined on 1 May 2017 from a genesis block with no allocation to anyone; the entire supply has been created as block rewards. The mining reward is a function of block height, computed by the protocol and not read from the coinbase. In the current era the reward declines approximately linearly with height toward a permanent floor of 0.5 BIS per block, which it does not fall below: the reward schedule has a floor, not a terminal zero, so block production continues to issue BIS indefinitely. In addition, on every tenth block up to block 4,380,000 the protocol paid a development reward equal to that block's mining reward and a separate, declining hypernode reward to two fixed, protocol-designated addresses — the development reward to the project's genesis address and the hypernode reward to a dedicated hypernode-payout address. Block 4,380,000 is the HF4 boundary at which the HyperNode metrics layer was retired, and the chain has passed it: those periodic payments have ended, and issuance now consists of the mining reward — which continues to decline toward its 0.5 BIS floor — and transaction fees.

The consequence, stated plainly, is that BIS has no fixed maximum supply. Because the per-block mining reward floors at a positive value rather than reaching zero, total supply increases without an upper bound and continues indefinitely: the per-block reward never falls below 0.5 BIS, so continued issuance dilutes existing holders. No inference about scarcity or future value should be drawn from the declining proportional rate of issuance. This is a material characteristic of the asset and is disclosed as such in Part F and Part G. Earlier project communications described a lower, bounded supply target; the figures in this document follow the emission rules as implemented in the protocol source, which is authoritative.

The smallest unit is 10^{-8} BIS. Reward, development, and hypernode payments and all balances are denominated in that unit, and no mechanism in the protocol increases or decreases an individual holder's balance other than a transaction the holder authorizes, a fee they pay, or a reward they mine.

Crypto-asset disclosures (MiCA Annex I)

The following parts present the disclosures listed in Annex I to Regulation (EU) 2023/1114 for a crypto-asset other than an asset-referenced token or an e-money token. Because Bismuth is fully decentralized and has no issuer and no offeror (see the *Regulatory positioning* note in the front matter and Section 22), items that presuppose such a person are completed by stating that the role does not exist, rather than by attributing it to the Bismuth Foundation. Where a disclosure would, on a formal admission to trading, belong to the person seeking admission or to the operator of the trading platform, that is stated. Because this crypto-asset white paper is drawn up by a person other than an offeror, an issuer, or an operator of a trading platform, it identifies the person that drew it up — the Bismuth Foundation — and the reason it did so, as required by Article 6(1), third subparagraph, of Regulation (EU) 2023/1114 (Part A and Section 22).

Part A — Person responsible for the crypto-asset white paper

1. **Name.** The Bismuth Foundation, the open-source steward of the Bismuth protocol and the preparer of this document.
2. **Legal form.** The Bismuth Foundation is an open-source developer collective, not a commercial issuer of BIS. It has not been constituted as a regulated entity for the purpose of a MiCA filing, because it is neither the issuer nor an offeror of BIS and makes no MiCA filing by publishing this document (Section 22).
3. **Registered address and head office.** Not constituted for a MiCA filing. On any formal admission of BIS to trading, the registered particulars are those of the person seeking admission or the operator that draws up the white paper.
4. **Date of registration.** Not applicable for the reason in item 2.
5. **Legal entity identifier.** The Bismuth Foundation does not hold an LEI for this purpose; none is required of a development steward that is neither issuer nor offeror.
6. **Contact.** Website `www.bismuth.cz`; public community channels are listed there. A monitored email address and telephone number for regulatory correspondence, with a stated response period, would be designated by the responsible person on any formal admission to trading.
7. **Parent company.** None.
8. **Management body.** The protocol is maintained by an open group of contributors under public version control; there is no corporate management body. The attestation in the front matter is made by the Foundation in that capacity, as preparer of this document.
9. **Business activity.** Non-commercial development, maintenance, and documentation of the open-source Bismuth protocol and reference node.
10. **Financial condition.** The Bismuth Foundation does not raise, hold, or manage purchaser funds for BIS and does not conduct an offer of BIS; it produces no offer-related financial statements. No financial-condition review under this item is applicable, because BIS is created only as a mining reward and is not sold by any person.

Part B — Issuer, if different from the person responsible

Not applicable. Bismuth has no issuer. BIS is created in a decentralized manner as a reward for validating transactions and maintaining the distributed ledger; no person has control over its creation, and there is no entity that issues BIS.

Part C — Operator of the trading platform

Not applicable to this document. This crypto-asset white paper is not drawn up by an operator of a trading platform. Where an operator admits BIS to trading on its own initiative without a white paper having been published, that operator must draw up the white paper (Article 5(3)) and complete this part with its own name, legal form, registered address, date of registration, legal entity identifier, parent company (if any), the reason it drew up the white paper, the identity and functions of its management body, and its business activity.

Part D — The crypto-asset project

1. **Name and ticker.** Project: Bismuth. Crypto-asset: BIS.
2. **Description.** Bismuth is a public, permissionless, proof-of-work blockchain with an account-based ledger and a generic, extensible transaction model, first mined on 1 May 2017. BIS is its native crypto-asset, used to transfer value and to pay transaction fees, and it is the unit that the protocol's virtual machine custodies. Sections 1–17 describe the project in technical detail.
3. **Persons involved.** The protocol is developed and maintained by the Bismuth Foundation and an open community of contributors under public version control at the project repository (Reference [6]). No advisor, placement agent, or crypto-asset service provider has been engaged in connection with an offer, because no offer is made.
4. **Utility-token features.** Not applicable. BIS is not a utility token and is not issued against the promise of any good or service.
5. **Milestones.** Bismuth's mainnet was first mined on 1 May 2017 as a fair launch — no pre-mine and no initial coin offering — and was described at the time as the first blockchain written in Python. Its difficulty controller and block-validation method were analysed in the peer-reviewed control-systems literature in 2017–2018 [2, 3, 13]. A plugin "building-blocks" architecture and a documented corpus of on-chain protocols (tokens, aliases, naming, social, event sourcing) followed in 2018, alongside a design essay separating consensus over data from consensus over meaning [14]; a HyperNode proof-of-stake metrics layer operated from around block 800,000. Later milestones include confidential (shielded) tokens and the first on-chain governance vote (2019), on-chain recording of real-world telemetry (2020), and cross-chain bridges to Ethereum and BNB Chain (2021). The HyperNode layer and the periodic development and hypernode rewards were retired at block 4,380,000 (the HF4 boundary), which the chain has now passed (Section 17). The current milestone is the coordinated *hf2* upgrade described in this paper, which is miner-activated and, as of this edition, has not yet activated on the public network (Section 16). The project is developed as open source and has raised no funds.
6. **Use of funds collected.** Not applicable. No funds or other crypto-assets have been collected from the public in connection with BIS.

Part E — Admission to trading

1. This crypto-asset white paper concerns the **admission to trading** and secondary-market circulation of an existing, freely created crypto-asset. It does not concern, and does not constitute, an offer of BIS to the public.
2. **Reasons.** BIS already circulates and is traded on third-party platforms outside the control of any single person. This document is published voluntarily to provide holders, prospective

holders, and any person who may seek admission of BIS to trading with the structured, comparable, risk-aware information that MiCA prescribes.

3. Amount to be raised, target subscription, and treatment of oversubscription: *not applicable* — no offer.
4. Issue price, subscription fee, or pricing method: *not applicable* — no offer; the market price of BIS is determined by third-party platforms.
5. Total number of crypto-assets offered or admitted: *not applicable* as an offered quantity; the supply schedule is described in item 8 of Part G and in Section 17.
6. **Prospective holders.** BIS is permissionless; there is no targeting of, or restriction on, holders at the protocol level. Eligibility to acquire BIS on a given platform is subject to that platform's and the holder's jurisdiction's rules.
7. Reimbursement on unmet target, withdrawal, or cancellation: *not applicable* — no offer and no purchaser funds are handled.
8. Phases of the offer and differential pricing: *not applicable*.
9. Subscription period: *not applicable*.
10. Safeguarding of funds and crypto-assets under Article 10: *not applicable* — no funds or crypto-assets are collected or safeguarded by any person in connection with BIS.
11. Methods of payment and of transfer of value for reimbursements: *not applicable*.
12. **Right of withdrawal (Article 13).** Not applicable. No offer to the public is made, and BIS is already admitted to trading; under Article 13(4) no 14-day right of withdrawal arises on secondary-market purchases.
13. **Transfer to holders.** BIS is transferred peer-to-peer on the Bismuth network. It is acquired by mining or through third-party platforms, and is delivered on-chain to the holder's address on confirmation.
14. **Technical requirements.** To hold BIS, a holder needs a Bismuth-compatible wallet controlling a Bismuth address (Section 7) and the ability to receive transactions on the network or to withdraw from a platform.
15. Crypto-asset service provider in charge of placement: *not applicable* — no placement.
16. **Trading platform.** BIS trades on third-party platforms; market-data listings are referenced from the project website. This document does not itself seek admission on a named MiCA-authorized venue; on any such admission, the responsible person names the platform, the access method, and the costs.
17. Expenses of the offer: *not applicable*.
18. **Conflicts of interest.** Contributors to the Bismuth project may hold BIS and may mine it; they receive no allocation other than mining and, until block 4,380,000, the protocol's development reward, which was paid to a fixed network address by the protocol (Section 17), not by any offer; that periodic payment has since ended.
19. **Applicable law and competent court.** The protocol is operated by a permissionless global network and is not governed by the law of any single jurisdiction. This voluntary document designates no governing law or competent court; on a formal admission to trading, the responsible person designates these.

Part F — The crypto-asset

1. **Type.** BIS is a crypto-asset other than an asset-referenced token or an e-money token. It is the native coin of a permissionless proof-of-work blockchain. It is not an electronic money token: it is not pegged to, and is not redeemable at par against, any official currency. It is not an asset-referenced token: it references no currency, asset, or basket and seeks to maintain no stable value.
2. **Characteristics, classification data, and functionality.** BIS is fungible and divisible to eight decimal places. It is freely transferable on-chain, confers no claim or right against any person (Part G), and is used to transfer value and pay transaction fees; it is also the unit of value that the reference virtual machine custodies (Section 10). For the classification of this white paper in the register kept under Article 109 (in the form specified by Commission Delegated Regulation (EU) 2025/421), the relevant data are: crypto-asset category — *other crypto-asset* (not an ART, not an EMT); distributed-ledger technology — the Bismuth proof-of-work blockchain; consensus mechanism — proof of work (Section 3); native to its own ledger (not a token issued on a third-party ledger). Functionality timing: value transfer and fee payment function today; the programmable (virtual-machine) and shielded functionalities, together with the other protocol changes bundled into the hf2 upgrade, are implemented and activate with the *hf2* boundary (full list and current status in Section 16).

Part G — Rights and obligations

1. **Rights and obligations of the holder.** None are attached to BIS. BIS confers no claim against the Bismuth Foundation or any other person, and no right to income, dividend, interest, redemption, repayment, profit-sharing, or governance. A holder may hold and transfer BIS; the protocol imposes no obligation on the holder beyond paying the transaction fee on a transaction the holder makes.
2. **Modification of rights and obligations.** There are no attached rights to modify. The rules under which BIS is created and transferred are the consensus rules of the protocol, which change only through the miner-activated upgrade process described in Section 16, identically for all holders, and not by the decision of any person.
3. Future offers by, or crypto-assets retained by, an issuer: *not applicable* — no issuer; no person retains an allocation of BIS.
4. Quality and quantity of goods or services for a utility token: *not applicable*.
5. Redemption of a utility token: *not applicable*.
6. **How BIS is acquired and sold.** BIS is acquired by mining or on third-party platforms and is transferred peer-to-peer on the Bismuth network between addresses (Section 7).
7. **Transferability.** The protocol imposes no restriction on the transfer of BIS; it is freely transferable on-chain. Restrictions may be imposed by the platforms on which BIS trades or by a holder's jurisdiction.
8. **Supply protocol.** BIS has no fixed maximum supply. New BIS is created only as a block reward, computed by the protocol from the block height. The mining reward declines approximately linearly with height to a permanent floor of 0.5 BIS per block, below which it does not fall; on every tenth block until block 4,380,000 a development reward and a separate, declining hypernode reward were paid to fixed network addresses, after which those periodic payments ceased. Because the mining reward floors at a positive value rather than reaching

zero, total supply increases without an upper bound and continues indefinitely; the per-block reward never falls below 0.5 BIS, so continued issuance dilutes existing holders, and no inference about scarcity or future value should be drawn from the declining proportional rate of issuance. There is no burn, rebase, or demand-responsive supply mechanism. Section 17 sets out the schedule.

9. Protection and compensation schemes: *none*. BIS is not covered by any value-protection or compensation scheme, nor by the investor compensation schemes under Directive 97/9/EC or the deposit guarantee schemes under Directive 2014/49/EU.
10. **Applicable law and competent court.** As in Part E, item 19: the permissionless protocol is not governed by the law of a single jurisdiction; this voluntary document designates none.

Part H — Underlying technology

1. **Technology and standards.** Bismuth is a permissionless distributed ledger maintained by a reference node written in Python (3.8 and later). The ledger is account-based; transactions are the eight-field records of Section 2. Cryptographic primitives include SHA-224 and BLAKE2b hashing, RSA (PKCS #1 v1.5), secp256k1 and NIST P-256 ECDSA, Ed25519, and the lattice-based ML-DSA family (FIPS 204); wallets use BIP32/BIP39 derivation. Consensus byte formats are frozen and characterization-locked (Sections 6–7). The reference implementation is open-source under the GNU General Public License.
2. **Consensus mechanism.** Proof of work. Blocks are produced by the memory-hard Heavy3 function described in Section 3, targeting a sixty-second interval, with difficulty retargeting per Section 4 and longest-valid-chain selection per Section 5. Proof of work is the consensus mechanism whose environmental impacts are disclosed in Section 18.
3. **Incentive mechanism and fees.** Miners are incentivized by the block reward (Section 17) and by transaction fees (Section 9). Fees are paid by the sender and accrue to the miner; the live fee is a small base fee plus a charge proportional to payload size.
4. **Operation of the ledger.** The distributed ledger is not operated by the Bismuth Foundation, by an issuer, or by any third party acting on their behalf. It is maintained by independent, permissionless nodes and miners that anyone may run; no person controls the network or can transfer, freeze, or reverse a holder's BIS.
5. **Audit of the technology.** The project applies replay verification (re-hashing the chain through the frozen serialization boundary), consensus-invariant tests, and characterization locks as continuous internal controls, and has conducted a documented internal adversarial security review of the modernization work, which identified and fixed specific defects (Reference [7]). The chain's difficulty-control and block-validation methods were, in addition, formally analysed in the peer-reviewed control-systems literature [2, 3, 13], and that difficulty-control work is cited among the prior art of IBM patent US 10,880,073 [15]; this is academic and prior-art scrutiny of specific mechanisms, not a security certification of the protocol as a whole. No external, independent third-party security certification of the protocol is claimed, and a holder should not infer one.

Part I — Risks

1. **Risks of the admission to trading.** BIS trades on third-party platforms whose pricing, liquidity, solvency, custody, and continued operation are outside the control of the Bismuth Foundation and of any person preparing this document. A platform may suspend trading, delist

BIS, fail, or be compromised, and the price formed on a platform may not reflect any fundamental value.

2. **Risks of the person responsible.** There is no issuer. The Bismuth Foundation is a voluntary, open-source developer group rather than a capitalized undertaking; it could reduce or cease its activity, and it gives no warranty and assumes no liability for the value or function of BIS. This concentration of maintenance in a small group is mitigated by the protocol being open-source and permissionlessly forkable, so that development can be continued or replaced by others.
3. **Risks of the crypto-asset.** BIS is highly volatile and may lose its value in part or in full; it may be illiquid, particularly as a small-capitalization asset, and may not always be transferable. It confers no claim, income, or redemption right, and is not covered by any investor-compensation or deposit-guarantee scheme. Its supply has no fixed maximum (Part G, item 8), so continued issuance exerts ongoing dilution. A holder who loses control of a private key loses the associated BIS irrecoverably, and on-chain transactions are irreversible once confirmed. The tax and regulatory treatment of acquiring, holding, and transferring BIS depends on the holder's jurisdiction and may change, and the regulatory characterisation of BIS and the standing of this document could be assessed differently by a competent authority.
4. **Risks of project implementation.** The coordinated *hf2* upgrade is miner-activated and, as of the date of this document, has not activated on the public network; the features it carries (the virtual machine, the shielded layer, BLAKE2b proof of work, and others in Section 16) are inert on the public network until then. Activation depends on miner adoption and may be delayed, may not occur, or may surface defects on activation. These risks are mitigated by gating each change behind a single deterministic on-chain activation height, by replay-verifying every storage and serialization change to be byte-identical across the boundary, and by the consensus-invariant and characterization test suites; nevertheless, software of this complexity may contain defects.
5. **Risks of the technology, and mitigations.** As a proof-of-work network, Bismuth is exposed to a majority-hashrate ("51%") attack, in which an entity controlling most mining power could reorganize recent blocks and attempt double-spends; the memory-hard Heavy3 function raises the cost of dedicated hardware, and reward maturity (activating with *hf2*) and waiting for confirmations reduce the impact of shallow reorganizations, but the risk is material for a smaller network and mining power may concentrate. Settlement is probabilistic, not final, until a transaction is buried under sufficient blocks. Consensus or implementation defects could cause chain splits or incorrect validation; these are mitigated by the frozen, characterization-locked serialization boundary, replay verification, value-conservation invariant tests, and — with *hf2* — a consensus-committed virtual-machine state root, but cannot be excluded. Classical RSA and elliptic-curve signatures would be vulnerable to a sufficiently capable quantum computer; the signer abstraction already supports the lattice-based ML-DSA family, so a migration is a dispatch-and-encoding change rather than a redesign (Section 15), though no migration has been made. Bootstrap snapshots are verified forward from a checksummed boundary rather than trusted blindly. Holders bear key-management and wallet-software risk.

18. Principal adverse environmental impacts

Article 6(1)(j) of Regulation (EU) 2023/1114 requires information on the principal adverse impacts on the climate and other environment-related adverse impacts of the consensus mechanism used to create the crypto-asset, in the form specified by Commission Delegated Regulation (EU) 2025/422. Bismuth uses an energy-intensive proof-of-work consensus mechanism (Section 3), so these impacts are a material disclosure.

For a permissionless network the exact energy and emissions figures are not directly measurable: the number, location, and efficiency of the machines that mine Bismuth are not known to any party. The values below are therefore **modelled estimates**, presented with the assumptions and method used to derive them, in the indicator structure of the Delegated Regulation. They are illustrative and should be refined with measured inputs; on a formal admission to trading the responsible person must publish the indicators with its own methodology and keep them current on a website (Article 6(1)(j); Section 20).

Table 1. Mandatory sustainability indicators (Commission Delegated Regulation (EU) 2025/422), modelled estimate. The 0.25 MW aggregate mining power below is an unverified illustrative assumption, not a measurement: the network's actual power draw is unknown and could be materially higher or lower, and the figures should be read as an order-of-magnitude model, not a precise account. Modelling basis: an assumed aggregate mining power draw of 0.25 MW; continuous operation (8,760 h/year); 525,600 blocks per year at the 60-second target; an assumed average of two economic transactions per block (1.05 million per year); an assumed grid carbon intensity of 0.45 kg CO2e/kWh location-based; and an assumed renewable share of 30%. Scope 1 is negligible because mining nodes consume purchased electricity rather than combusting fuel on site.

ID	Indicator	Unit	Modelled value
S.8	Energy consumption (validation and ledger maintenance)	kWh / year	2,190,000
S.10	Renewable energy consumption (share of total)	%	30.0
S.11	Energy intensity per validated transaction	kWh / tx	2.09
S.12	Scope 1 GHG emissions	tCO2e / year	≈ 0
S.13	Scope 2 GHG emissions	tCO2e / year	986
S.14	GHG intensity per validated transaction	kg CO2e / tx	0.94

Definitions follow the Delegated Regulation: "energy consumption" is the total energy used for transaction validation and maintenance of ledger integrity; "renewable energy" is as defined in Directive (EU) 2018/2001; greenhouse-gas emissions are expressed in tonnes of CO2-equivalent; Scope 1 covers emissions from sources controlled by the network's nodes and Scope 2 covers emissions from the generation of the electricity they purchase. The estimates scale linearly with the assumed mining power and grid intensity and inversely with the assumed transaction count; a reader can recompute them with better inputs. The project does not operate the mining hardware and therefore cannot commit other parties to energy-reduction targets; the structural mitigation is the memory-hard design (Section 3), which favours commodity hardware over dedicated high-power machines, and the modest absolute scale of a small network's aggregate mining power.

19. Marketing communications

Article 7 of Regulation (EU) 2023/1114 governs marketing communications relating to a crypto-asset. Any marketing communication concerning BIS that is disseminated by a person within the scope of MiCA must be clearly identifiable as a marketing communication; must be fair, clear, and not misleading; must be consistent with this crypto-asset white paper; must state that a crypto-asset white paper has been published and indicate the website address of the responsible person together with a contact telephone number and email address; and must carry the prescribed statement. No marketing communication may be disseminated before the white paper is published. The prescribed statement, with the responsible-party label adapted to the actual disseminator, reads:

> **STATEMENT UNDER ARTICLE 7(1)(E)**
>
> This crypto-asset marketing communication has not been reviewed or approved by any competent authority in any Member State of the European Union. The person responsible for this crypto-asset marketing communication is solely responsible for the content of this crypto-asset marketing communication.

The Bismuth Foundation does not market BIS as an investment and makes no representation as to its future value (Article 6(4)). Because BIS has no issuer or offeror, MiCA's marketing obligations bind whichever person, if any, disseminates marketing communications about BIS within scope; this document provides the white paper with which such communications must be consistent.

20. Notification, publication, and modification

This section records how the obligations in Articles 8, 9, 12, and 13 of Regulation (EU) 2023/1114 relate to this document. **Notification (Article 8).** A person seeking admission of BIS to trading, or an operator admitting it, must notify the crypto-asset white paper to the competent authority of its home Member State at least 20 working days before publication. This voluntary document is not such a notification and has not been notified to any competent authority; it carries no date of notification because none has occurred (the front matter states the date of the edition instead). **Publication (Article 9).** A required white paper must be published on a publicly accessible website before the start of admission to trading and kept available; this document is published on the project website (Reference [6]) and is freely accessible. **Modification (Article 12).** A required white paper must be modified whenever a significant new factor, material mistake, or material inaccuracy capable of affecting the assessment of BIS arises. The Bismuth Foundation maintains this document by edition: the authoritative source for the protocol's behaviour is its open-source code, and material protocol changes — in particular the activation of *hf2* (Section 16) — will be reflected in a revised edition, with the revision history kept at the end of the document. **Right of withdrawal (Article 13).** No right of withdrawal arises: no offer to the public is made, and BIS is already admitted to trading, so under Article 13(4) the 14-day withdrawal right does not apply to secondary-market purchases.

21. Machine-readable format

Article 6(10) of Regulation (EU) 2023/1114 requires the crypto-asset white paper to be made available in a machine-readable format. Commission Implementing Regulation (EU) 2024/2984 specifies that format as a single XHTML document in which the Annex I fields are marked up using the Inline XBRL (iXBRL) 1.1 specification and the ESMA MiCA reporting taxonomy, the relevant element set for a crypto-asset other than an asset-referenced or e-money token being that of Table 2 of the Annex to that Regulation; the obligation applies from 23 December 2025. This document is authored as a single, valid XHTML file with the disclosure parts structured to that template, so that the Inline XBRL tags required on a formal admission can be applied to the existing markup without restructuring the document. The human-readable rendering and the machine-readable instance are one file. On a formal admission to trading, the responsible person produces the iXBRL instance against the published ESMA taxonomy.

22. Standing of this document

This document is published voluntarily by the Bismuth Foundation on the template of Article 6 and Annex I of Regulation (EU) 2023/1114. It is drawn up so that holders, prospective holders, trading platforms, and competent authorities have the structured, comparable information that the Regulation is designed to provide. It is not a crypto-asset white paper that has been approved by, or notified to, any competent authority, and publishing it does not make the Bismuth Foundation an issuer, an offeror, or a person seeking admission to trading.

Bismuth is fully decentralized. Under Recital 22 and Article 4 of MiCA, the distribution of a crypto-asset that is created as a reward for maintaining the distributed ledger and is offered for free does not require a crypto-asset white paper; the white-paper obligation under Article 5 is engaged on admission to trading and falls on the person seeking admission or the operator of the trading platform (Article 5(3)), and Article 15 limits an operator's liability where no white paper can be produced because there is no offeror. Where this document uses the statements that MiCA prescribes — on the first page, in the management-body statement, and in the marketing-communication statement — it reproduces the prescribed wording and adapts only the label of the responsible person to reflect that the preparer is the Foundation and not an offeror. A person who seeks to admit BIS to trading, or who makes an offer of BIS, becomes the responsible person under MiCA and must satisfy the Regulation in full, including notification (Article 8), publication (Article 9), the machine-readable format (Section 21), and ongoing modification (Article 12). The ultimate source of truth for every technical statement in this document is the protocol's open-source code (Reference [6]); where an earlier project description and the code disagree, the code governs.

References

[1] S. Nakamoto, “Bitcoin: A Peer-to-Peer Electronic Cash System,” 2008.

[2] G. Hovland and J. Kučera, “Nonlinear Feedback Control and Stability Analysis of a Proof-of-Work Blockchain,” *Modeling, Identification and Control*, vol. 38, no. 4, pp. 157–168, 2017. doi:10.4173/mic.2017.4.1.

[3] J. Kučera and G. Hovland, “Tail Removal Block Validation: Implementation and Analysis,” *Modeling, Identification and Control*, vol. 39, no. 3, pp. 151–156, 2018. doi:10.4173/mic.2018.3.1.

[4] National Institute of Standards and Technology, “Module-Lattice-Based Digital Signature Standard,” FIPS 204, 2024.

[5] A. Back, “Hashcash — A Denial of Service Counter-Measure,” 2002.

[6] The Bismuth Foundation, reference node and protocol source, *github.com/hclivess/Bismuth* and *www.bismuth.cz*.

[7] The Bismuth Foundation, “Bismuth: Protocol and Node Architecture,” v2.0, 2026, and the protocol notes in the project repository (including the security-review and hard-fork notes).

[8] Regulation (EU) 2023/1114 of the European Parliament and of the Council of 31 May 2023 on markets in crypto-assets (MiCA), OJ L 150, 9.6.2023.

[9] Commission Implementing Regulation (EU) 2024/2984 of 29 November 2024 laying down implementing technical standards for the application of Regulation (EU) 2023/1114 as regards standard forms, formats and templates for crypto-asset white papers, OJ, 3.12.2024.

[10] Commission Delegated Regulation (EU) 2025/421 (regulatory technical standards on the data necessary for the classification of crypto-asset white papers, Article 109(8)).

[11] Commission Delegated Regulation (EU) 2025/422 (regulatory technical standards on the content, methodologies and presentation of information on principal adverse impacts on the climate and other environment-related adverse impacts of the consensus mechanism).

[12] M. Palatinus, P. Rusnak, A. Voisine, and S. Bowe, “Mnemonic code for generating deterministic keys,” BIP 39, and P. Wuille, “Hierarchical Deterministic Wallets,” BIP 32.

[13] G. Hovland and J. Kučera, “Security of the Bismuth Blockchain,” 2018.

[14] J. Kučera, “Semantic Interpretation,” 19 July 2018 — project essay on separating consensus over data from consensus over meaning.

[15] Hwang et al. (assignee International Business Machines Corporation), “Optimizing Performance of a Blockchain,” U.S. Patent 10,880,073 B2, granted 29 December 2020, which cites Reference [2] among its “Other Publications” prior art.

[16] The Bismuth Foundation, “10 Years of Bismuth: Architectural Anticipation in a Fair-Launched Blockchain, 2017–2026,” and the project’s interview and coverage record at *www.bismuth.cz* — primary and self-published sources, provided for background and not as independent verification.

Revision history

- Edition 2026.1, 28 June 2026: first edition drawn up on the template of Regulation (EU) 2023/1114, Article 6 and Annex I. Rewritten from the protocol source as a technical and disclosure document; supersedes the 2019 Bismuth white paper.